Anubhav Dogra

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WORK EXPERIENCE

Volunteer Software Developer · Translation Commons · California — April 2025 - February 2026

- Standardized and cleansed multi-source linguistic datasets using Python, Pandas, and SQL, improving data completeness and consistency across repositories.
- Built interactive Tableau dashboards for organizational leadership to monitor volunteer onboarding, project completion, and progress.

Data Analyst Intern · Steel Authority of India Limited — March 2022 - August 2022

- Designed and maintained 3 interactive Tableau dashboards used by 20+ stakeholders to monitor five key operational KPIs and support data-driven decision-making.

Data Analyst Intern · Steel Authority of India Limited — March 2021 - August 2021

- Conducted exploratory data analysis (EDA) on five years of historical data, identifying seasonal trends that informed a 15% budget reallocation.

EDUCATION

The State University of New York at Albany · MS Business Analytics — December 2024

Coursework: Predictive Analytics, Statistical Analysis, Marketing Analytics, Data Mining, Databases, Forecasting, Panel Data Analysis, Visualization

Chhattisgarh Swami Vivekanand Technical University · B. Tech. Electronics & Telecommunication Engineering — December 2020

SKILLS

AI & LLM Engineering: Python, OpenAI, Anthropic Claude, Google Gemini, Gemini Enterprise Agent Platform, LangChain, LangGraph, Google ADK, NVIDIA NAT, BeeAI, Aisuite, Generative AI, Hugging Face

Agentic AI & LLM Architecture: Multi-Agent Systems, Agent Orchestration, ReAct, Reflection, Planning, Tool Calling, Function Calling, RAG, MCP, A2A Protocol, Memory Systems (Semantic/Episodic/Procedural), Guardrails, Prompt Engineering, Fine-Tuning, Knowledge Graphs

AI Evaluation & Observability: End-to-End Evals, Component-Level Evals, Arize Phoenix, OpenInference, OpenTelemetry, Ragas, LLM-as-a-Judge, LLM Evaluation

Backend & APIs: FastAPI, Flask, REST APIs, AsyncIO, Pydantic, SQLAlchemy, Node.js, TypeScript

Databases & Vector Stores: PostgreSQL, MongoDB, SQLite, DuckDB, TinyDB, ChromaDB, Weaviate, DocArray

Data Science & Machine Learning: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, SQL

Analytics & BI: Tableau, Power BI, Microsoft Excel, Google Sheets, Looker, Google Analytics, Adobe Analytics, Meta Analytics, Marketo

Cloud & Data Platforms: AWS, GCP, Azure, Snowflake, Databricks

CERTIFICATIONS

Agentic AI, RAG, Pretraining LLMs, MCP, A2A Protocol, LangChain for LLM App Development, LangChain for AI Agents, LangGraph for AI Agents, Nvidia NeMo Agent Toolkit, LangChain for Data, Long-Term Agentic Memory with LangGraph, Guardrails, Marketing Analytics, Data Analytics, Agile, Tableau BI Analyst, Sales CRM, Digital Marketing

PORTFOLIO PROJECTS

Retail Demand Forecasting · Live demo · GitHub — October 2026

Python, scikit-learn, pandas, Streamlit, Claude API

- Built an end-to-end demand forecasting system on ~1M retail transactions: data cleaning, feature engineering, and pooled gradient-boosting models with median and 90th-percentile forecasts, evaluated with a leakage-tested rolling backtest (8 folds × 4 horizons).
- Reduced forecast error ~10% versus a 4-week moving average (WAPE 0.537 vs 0.595); in a stock-planning simulation, the model policy left ~14% fewer unsold units at the same 88.8% fill rate.
- Deployed an interactive Streamlit dashboard with a Claude-powered assistant that explains forecasts from computed facts only; added unit tests and per-session API usage limits.

Healthcare Concierge Multi-Agent System · GitHub — August 2026

Python, BeeAI, LangChain, Claude, Gemini, FastMCP, A2A Protocol, Google ADK, Arize Phoenix

- Architected a distributed multi-agent system using BeeAI and the Google A2A protocol to coordinate 4 microservices across distinct network ports.
- Built hybrid LLM pipelines with Gemini 3.1 Flash Lite for orchestration and medical research, and Claude Haiku 4.5 for zero-shot PDF policy grounding.
- Integrated Model Context Protocol (FastMCP) via langchain-mcp-adapters for structured provider lookups using stdio transport.
- Implemented full-stack observability with Arize Phoenix and OpenInference to trace agent trajectories, tool calls, and token usage.

Climate Intelligence & Multi-Agent Analysis Platform · GitHub — August 2026

Python, NVIDIA NeMo Agent Toolkit (NAT), LangGraph, Llama 3.1 70B (NIM), Arize Phoenix, Ragas, FastAPI, Next.js

- Built a multi-tool ReAct agent with NVIDIA NAT and Llama 3.1 70B NIM to query NOAA datasets, generate visualizations, and compute decadal climate statistics using async Pydantic schemas.
- Integrated a specialized LangGraph calculator agent via the LLM-lifting pattern to execute complex multi-step math (CAGR, exponential projections) without hardcoding models.
- Implemented OpenTelemetry tracing with Arize Phoenix, diagnosing tool-selection bottlenecks to cut tool calls by 70% and response latency by 60%.
- Designed ground-truth evaluation pipelines using Ragas (LLM-as-a-judge), resolving silent data-filtering bugs to raise answer accuracy from 0.0 to 1.0 (100%).
- Exposed the agent as an OpenAI-compatible REST API (nat serve) integrated with a real-time streaming Next.js/React interface.

Autonomous Multi-Agent Marketing Pipeline · GitHub — August 2026

Python, Aisuite, LangGraph, OpenAI (o4-mini, gpt-image-1-mini), Tavily API

- Built a stateful 4-agent pipeline using LangGraph (StateGraph) with centralized TypedDict state management to automate campaign creation end-to-end.
- Integrated Tavily Web Search and internal catalog tools to ground trend analysis in real-time market data and live inventory.
- Chained gpt-image-1-mini for visual asset synthesis with multimodal o4-mini prompting to generate context-aware marketing copy.
- Programmed an agent to compile cross-agent outputs into executive-ready Markdown reports with strategic justifications and inventory recommendations.

Multi-Server AI Research Assistant (MCP) · GitHub — July 2026

Python, FastMCP, Anthropic Claude API, AsyncIO, arXiv API

- Built an MCP client-server ecosystem using Anthropic's Claude to automate academic research, data extraction, and synthesis.
- Developed a FastMCP research server exposing tools for arXiv search, dynamic resource endpoints (papers://{topic}), and structured prompt templates.
- Multiplexed multiple MCP servers (Research, fetch, filesystem) using AsyncExitStack for concurrent tool and resource handling.
- Deployed dual-transport architecture supporting local execution via STDIO and remote containerized endpoints via SSE.

Autonomous Multi-Agent Research System · GitHub July 2026

Python, LangGraph, OpenAI, Tavily, Gradio, SQLite

- Built an AI essay writing system using LangGraph and OpenAI (gpt-3.5-turbo) to automate essay planning, research, drafting, and reflection
- Integrated TavilyClient to execute real-time web searches based on generated research plans and feedback critiques, ensuring data-backed draft updates.
- Engineered conditional routing logic (should_continue) to auto-evaluate revision limits and guide execution flow between reflection loops and graph completion.
- Extended the graph with LangGraph SQLite checkpointers (SqliteSaver) to implement Human-in-the-Loop (HITL) workflows, persisting state to disk to pause execution for manual outline or draft approval before re-entering revision cycles

RAG-Based E-Commerce Support Chatbot · GitHub July 2026

Python, Weaviate, Flask, LangChain

- Designed and built a Retrieval-Augmented Generation chatbot for e-commerce support, combining LLM-based query routing, vector search over product/FAQ data (Weaviate), and structured filter extraction for natural-language product search.
- Reduced per-query cost and latency through prompt optimization and added distributed tracing (Arize Phoenix/OpenTelemetry) for production observability.

Autonomous Multi-Agent AI System with Tool Orchestration & Reflection · GitHub June 2026

Python, Aisuite, OpenAI API, Anthropic API, FastAPI, SQLite, SQLAlchemy, TinyDB, DuckDB, Matplotlib, Tavily API, arXiv API, Wikipedia API

- Built an end-to-end multi-agent system leveraging ReAct, Reflection, and Planning-in-Code patterns to automate complex research and workflow execution.
- Created an iterative multimodal reflection pipeline (GPT-4o, Claude 3.5 Sonnet) that autonomously critiques and refines generated Matplotlib charts.
- Designed a 4-agent LangGraph workflow (Research, Design, Copywriting, Packaging) grounded with live web search (Tavily) and local product databases.
- Implemented component-level evaluations using domain allow-lists and LLM-as-a-Judge rubrics to benchmark retrieval quality.

Retrieval-Augmented Generation (RAG) Chatbot · GitHub June 2026

Python, LangChain, OpenAI API, ChromaDB, Panel, Whisper

- Built a conversational chatbot with a Panel-based UI, utilizing LangChain and ConversationBufferMemory for context-aware, multi-turn Q&A on custom datasets.
- Engineered an ingestion pipeline for PDFs, Notion databases, and Whisper-transcribed audio, applying context-aware text chunking and persisting OpenAI embeddings in ChromaDB.
- Optimized search accuracy, relevance, and diversity by implementing Maximum Marginal Relevance (MMR), LLM-powered self-querying filters, and contextual compression.

AI Subreddit Automation Bot · Live demo May 2026

TypeScript, Node.js, Redis, Google Gemini Flash, Reddit Devvit API

- Deployed a 24/7 cloud-native AI agent installed on active subreddits (r/AITechBusiness) to autonomously monitor new posts and generate contextual replies using Google Gemini Flash.

Medical Appointment No-Show Prediction · GitHub June 2025

Python, Matplotlib, Pandas, NumPy, Seaborn, Excel, CSV

- Analyzed 110,000 public-health appointment records & built a Histogram-based Gradient Boosting classifier tuned to flag high-risk visits for targeted SMS/call reminders or overbooking. Identified booking-to-visit lead time as the top predictor – no-show rate climbs from ~5% same-day to ~30%+ in 30+ days.

Superstore Sales & Profitability Dashboard · Tableau February 2025

Tableau, Data Storytelling, Excel, CSV

- Built an interactive Tableau Story tracking sales, profit margins, and regional performance across retail product categories.
- Implemented dynamic filters, calculated metrics, and trend charts to identify profit leakage and high-performing segments.

Business Requirements Document Analysis · GitHub January 2025

Google Docs, Diagrams.net

- Developed functional requirements, including a process flow diagram and a domain model. Applied business analysis techniques such as requirements elicitation, gap analysis, and workflow modeling.